Algebra II Math Notes

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Contents

Chapter 1

Conic Sections

This chapter covers circles, ellipses, parabolas, and hyperbolas.

1.1 Conic Sections

When a double cone is sliced by a plane, the cross section is formed by the plane and the cone is called a **Conic Section**. The four main conic sections are the circle, the parabola, the ellipse, and the hyperbola.

1.1.1 Circle

A **circle** is a set of point which are equidistant from one point. The one point being called the **center** and the distance of the line is the **radius**. The standard form equation with the center at (0,0) is: $x^2 + y^2 = r^2$

1.1.2 Parabola

A **parabola** is the set of points in a plane that are the same distance from a given point and a given line in that plane.

The given point is the **focus** and the line is called the **directrix**.

The midpoint of the perpendicular segment from the focus to the directrix is called the **vertex** of the parabola.

The line which passes through the vertex and focus is called the **axis of symmetry**.

1.1.3 Ellipse

An **Ellipse** is the set of points in a plane such that the sum of the distances from two given points in that plane stays constant.

The two points are each called a **focus point** – plural *foci*.

The midpoint of the segment joining the foci is called the **center** of the ellipse.

Ellips have two axes of symmetry:
Longer one = **Major Axis**, Shorter one = **Minor Axis**.
The **major intercepts** are where the 2 points intercept with the major axis.
The **minor intercepts** are where the 2 points intercept with the minor axis.

1.1.4 Hyperbola

A **hyperbola** is the set of all points in a plane such that the absolute value of the difference of the distances between two given points stays constant.

Two given points are the **foci** of the hyperbola.

Hyperbolas looks like two opposing 'U-shaped' curves.

Hyperbolas have two axes of symmetry:
The one going through the hyperbola's foci is the **transverse axis.**
The one going through the center and is perpendicular to the transverse is the **conjugate axis**.

Chapter 2

System of Equations with Three Variables

Details methods (substitution and elimination) for solving systems of three linear equations

2.1 Linear Equations In Three Variables

2.1.1 Linear Equations: Solutions Using Elimination

Same as doing it for 2 equations (see previous chapter) except at the end you apply the found variables to any remaining equations.

2.1.2 Linear Equations: Solutions Using Matrices

Organized method of elimination and the same as with two variables (Chapter 2).

2.1.3 Linear Equations: Solutions Using Determinants

Remembering that determinants are organized by row and columns, unlike matrices,with three variables it stays the same. I recommend watching a video as this gets confusing quick.

2.1.4 Linear Equations: Solutions Using Substitution

Steps:

1. Select one equation to isolate a variable in. Example: Isolate y in E_1
2. Substitute what the variable equaled into the second equation. Example: y from E_1 into E_2
3. Solve the new E_2 now that it only has one variable to solve for.
4. Using both newly found variables, substitute them into both equations to make sure it's correct. Example result should be 1=1 -5=-5.

If the result is not equal like the example in step 4 then the system is inconsistent and there is no solution.

2.1.5 Linear Equations: Solutions Using Elimination

Steps:

1. Stack both equations while they are in *standard form*.
2. Multiply both equations so that one variable is eliminated by being opposites of eachother.
3. Add the equations – leaving one variable in the equation.
4. Solve for the remaining variable
5. Repeat steps 2-4 or solve for the second variable through substitution.
6. Check your solution by plugging in the variables into both original equations.

2.1.6 Linear Equations: Solutions Using Matrices

A **matrix** is a rectangular array of numbers or variables. It represents the system of equations in standard form using only coefficients and constants.

$$\begin{bmatrix} 2 & 4 & 6 \\ 1 & 2 & 3 \end{bmatrix}$$

The last column being what each equation equals. Now solve it the same as elimination.

2.1.7 Linear Equations: Solutions Using Determinants

A **determinant** is a square array of numbers or variables enclosed between vertical lines. They differ from matrices as determinants have a numerical value.

You solve by multiplying diagnonally and then doing the same again but this time using the original equations answers. This is also called **Cramer's Rule**.

2.1.8 Linear Inequalities: Solutions Using Graphing

You graph each equation and shade the section which makes the equation true. Afterwards there will be one specific section where they overlap. If they dont overlap, then there is no solution.

Chapter 3

Exponential and Logarithmic Functions

Explores exponential growth and decay, graphing exponential functions, defining and using logarithms, and solving equations involving exponents and logarithms.

3.1 Exponential and Logarithmic Functions

3.1.1 Exponential Functions

Any function defined by $y = b^x$, where $b > 0, b \neq 1$, and x is a real number, is called an exponential function.

3.1.2 Logarithmic Functions

if $x = 2^y$ were to be solved for y then you'd be finding the **logarithm**.

Common Logarithms are ones with base 10.
Natural Logarithms are ones with base e.

Chapter 4

Factoring Polynomials

4.1 Factoring Polynomials

Factoring is when you rewrite a polynomial as a product of polynomials.

4.1.1 Greatest Common Factor

The first method is called **greatest common factor** as you factor out the largest number in the polynomial.

Example: $5x + 5y = 5(x + y)$

4.1.2 Difference of Squares

The product of conjugates produces a pattern called a **difference of squares**.

Example: $(a + b)(a - b) = a^2 - b^2$

4.1.3 Sum or Difference of Cubes

Sum of Cubes is the form $(a)^3 + (b)^3$.
Difference of Cubes is the form of $(a)^3 - (b)^3$

They have similar factored patterns where they follow the same rules. The first binomial has the same sign, but the trinomial has the *opposite* sign.
$(a)^3 + (b)^3 = [(a) + (b)][(a)^2 - (a)(b) + (b)^2]$

$(a)^3 - (b)^3 = [(a) + (b)][(a)^2 + (a)(b) + (b)^2]$

Trinomials of the Form $x^2 + bx + c$ Example: $2x^2 + 16x + 24$

Step 1: See if you can take out the Greatest Common Factor to make the problem easier. In this case you can take out a *2* from all three numbers.

To factor these you first distribute the first x^2 into two parenthesis to start:
Don't forget to put the 2 you factored out in the beginning. So far we have $2(x^2 + 8x + 12)$. $2(x)(x)$

Now you think of two numbers which can add up the middle number but also multiply to the last number. Keep in mind you can switch up the symbols to make either one positive or negative. Some factors to try are: $(1)(12)$, $(-1)(-12)$, $(2)(6)$, $(3)(4)$

Plugging in 2 and 6 we can see that they add up to 8 and multiply to 12 giving us our factored trinomial, which keep in mind is just two binomial.

$2(x + 2)(x + 6)$

4.1.4 Sqaure Trinomials

Recall $(x + y)^2 = (x)^2 + 2(x)(y) + (y)^2$

Therefore, it is easy to see if an equation is a square trinomial or binomial.

Example: $4x^2 - 20x + 25$ can be deduced down to squares: $4x^2 = (2x)^2$ & $25 = (-5)^2$
Leaving us with the final step of putting them in the form $2xy$ to see if it fits: 2(2x)(-5) which does equal -20x like the original equation. This means $(2x - 5)^2$ is the square binomial to our original trinomial.

4.1.5 Factoring by Regrouping

There are two main things to look for in attempting to factor a polynomial of four or more terms with no GCF. It can be regrouped into two groups of two terms or two groups where one has three terms aka a trinomial.

Example: (az + ay) + (bz + by)

We can factor out both an 'a' and a 'b'. Leaving us with $a(y + z) + b(y + z)$. Now we see we can see that we can factor out (y + z) leaving us now with $(y + z)(a + b)$.

Chapter 5

Linear Equations

5.1 Linear equations

Linear Equation is one variable with the exponent 1 in an equation. Also known as first degree equations.
Formulas are equations that describes a relationship between unknown values.

$$Perimeter = 2l + 2w \tag{5.1}$$

$$Trapezoid = \frac{1}{2}h(b_1 + b_2) \tag{5.2}$$

$$RateDistance = \frac{D}{T} = R \tag{5.3}$$

$$AccumulatedInterest = \frac{A - P}{PR} = T \tag{5.4}$$

$$absolutevalue_1 = |A| + |-A| = A + A \tag{5.5}$$

5.1.1 Linear Inequalities

inequality is a sentence using symbols other than equality

5.1.2 Compound Equalities

Compound Inequality is a sentence with two inequality statements

Chapter 6

Linear Sentences In Two Variables

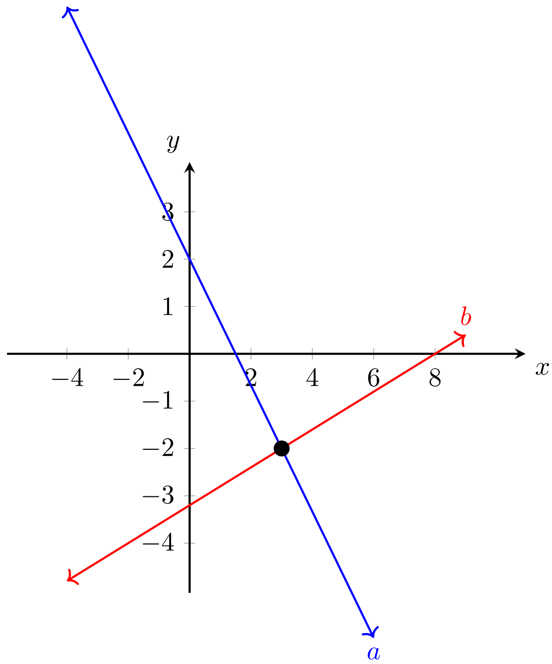
6.1 Linear Sentences In Two Variables

You can solve two equations with the same variables by graphing, substitution, elimination, matrices, anddeterminants.

6.1.1 Linear Equations: Solutions Using Graphing

To solve by graphing, you plot both equations on a graph and see where they intersect.

Example: $4x + 3y = 6$ & $2x - 5y = 16$



From the graph we can see they intersect at (3,-2)

6.1.2 Linear Equations: Solutions Using Substitution

Steps:

1. Select one equation to isolate a variable in. Example: Isolate y in E_1
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6.1.3 Linear Equations: Solutions Using Elimination

Steps:

1. Stack both equations while they are in *standard form*.
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6.1.6 Linear Inequalities: Solutions Using Graphing

You graph each equation and shade the section which makes the equation true. Afterwards there will be one specific section where they overlap. If they dont overlap, then there is no solution.

Chapter 7

Polynomial Arithmetic

7.1 Polynomial Arithmetic

7.1.1 Adding and Subtracting Polynomials

Monomial is an expression that could be a numeral, a variable, or the product of numerals and variables within certain restrictions.

1. Must have whole number exponents.
2. Variables do not appear under simplified radical expressions.
3. Denominators do not contain variables.

Examples: -10, a, $5t^2$, $3/8x^2y^3$

Binomial is an expression that is the sum of two monomials.
Trinomial is an expression that is the sum of three monomials.
Polynomial is an expression that is a monomial or the sum of two or more monomials.

7.1.2 Multiplying Polynomials

1. $x^a x^b = x^{a+b}$
Multiplying monomials with the same base you keep the base and add the powers.

2. $(x^a)^b = x^{ab}$
Keep the base and multiply the powers when finding powers of a base.

3. $(xy)^a = x^a y^a$
Raise each factor in the proudct to the power when finding the powers of a product

7.1.3 Special Products of Binomials

Conjugates are two binomials with the same two terms but opposite signs separating the terms. They are conjugates of each other.

Example: $3x + 2$ & $3x - 2$

Difference of Squares is the *product* of conjugates which produces a special pattern.

Example: $(x + y)(x - y) = x^2 - y^2$

Square Trinomial is the pattern produced by squaring a binomial.

Example: $(x + y)^2 = x^2 + 2xy + y^2$ & $(x - y)^2 = x^2 - 2xy + y^2$

Notice how only the *first* sign is changed when subtracting a squared binomial.

7.1.4 Dividing Polynomials

$$\boxed{a^m/a^n = a^{m-n}}$$

This works as long as a does not equal zero.

$$\boxed{a^-n = 1/a^n \text{ \& } 1/a^-n = a^n}$$

This works as long as a does not equal zero.
Also works in reverse.

7.1.5 synthetic Division

Synthetic Division is a shortcut for polynomial division when the divisor is the form of x-a. Only the coefficients are used when dividing.

Example: Divide $(3x^2 + 2x - 11)$ by $(x - 3)$

First step would be multiplying $(x-3)$ by $3x^2$ giving you $3x^3-9x^2$. You subtract this and keep repeating until you are left with a remainder. The remainder is added at the end by being divided by the divisor.

Chapter 8

Polynomial Functions

8.1 Polynomial Functions

A **Polynomial Function** is any function of the form:

$$a_0x^n + a_1x^{n-1} + a_2x^{n-2} + ... + a_{n-1}x + a_n$$

8.1.1 Zeros of a Function

Zeros of a Function is when you make a function equal zero so you can find the roots which equal zero. This tells you the x-intercepts of the graph.

Chapter 9

Radicals and Complex Numbers

9.1 Radicals and Complex Numbers

Radical Expression is the form $\sqrt[n]{a}$. The $\sqrt{}$ is a radical sign and the expression under it is the *radicand*, while the n is called the *index*.

$$\boxed{\text{If } x = \sqrt[n]{a}, \text{ then } x^n = a}$$

9.1.1 Adding and Subtracting Radical Expressions

Radical expressions are called **like radical expressions** if the indexes and radicands are the same.

Example: $3\sqrt{7} + 5\sqrt{7} = 8\sqrt{7}$

9.1.2 Multiplying Radical Expressions

To multiply we simply distribute each one then add them.

9.1.3 Dividing Radical Expressions

To divide we simply apply the sqrt to both top and bottom.

9.1.4 Rational Exponents

if n is a natural number greater than 1, and b is any nonnegative real number, then:

$$b^{1/n} = \sqrt[n]{(b)}$$

9.1.5 Complex Numbers

Imaginary values and pure imaginary numbers. The expression $\sqrt{-1}$ has no real answer. The symbol for it is i because its called an imaginary value. Any expression that is a product of a real number with i is called a pure imaginary number.

$$i = \sqrt{-1},\ i^2 = -1$$

Chapter 10

Rational Expressions

10.1 Rational Expressions

The quotient of two polynomials is called a **rational expression**, where the denominator is never zero.

Examples include $5/6$, $\frac{3x+8}{2x-9}$, $6x+5$

The last one can be written as $\frac{6x+5}{1}$ which fits the definition.

10.1.1 Simplifying Rational Expressions

1. Completely factor numerators and denominators. 2. Reduce common factors.

10.1.2 Multiplying Rational Expressions

1. Completely factor all numerators and denominators.
2. Reduce all common factors.
3. Either multiply the denominators, using FOIL if needed, or leave it in factored form.

10.1.3 Dividing Rational Expressions

Division of fractions involves multiplying by the *reciprocal* of the divisor.

10.1.4 Adding and Subtracting Rational Expressions

If they have the same denominator then add/subtract as usual making sure to simplify at the end.
If they *dont* have the same denominator then you need to find the Lowest Common Denominator (LCD).

10.1.5 Complex Fractions

If the numerator or denominator or both contain fractions the expression is called a **complex fraction.**

1. Simplify both numerator and denominator expressions into single fraction expressions.
2. Divide the numerator by the denominator and simplify if possible.

10.1.6 Proportion, Direct Variation, Inverse Variation, Joint Variation

Proportion is an equation stating that two rational expressions are equal. Simple ones are solved with the *Cross Product Rule*

Cross Product Rule
if $\frac{a}{b} = \frac{c}{d}$, then $ad = bc$.

Direct Variation is when 'y varies directly as x' or 'y is directly proportional to x'. This may be inteperted in two ways:

1. $\frac{y}{x} = k$

For some constant k - k is called the **constant of proportionality**. This is used when the constant is the desired result.

2. $\frac{y_1}{x_1} = \frac{y_2}{x_2}$

This is when the desired result is either an original or new value of x or y.

Inverse Variation is when 'y varies inversely (opposite) to x'.

Joint Variation is when one variable varies as the product of the other variables. The phrase goes 'y varies jointly as x and z':

1. $\frac{y}{xz} = k$

if the constant is desired.

2. $\frac{y_1}{x_1 z_1} = \frac{y_2}{x_2 z_2}$

if one of the variables is desired.

Chapter 11

Relations and Functions

This chapter talks about sets of ordered pairs and how they are symbolized and represented.

11.1 Relations And Functions

Relation is a set of ordered pairs that can be represented by a diagram, graph, or sentence.

The **Domain** is the set of all *first* numbers of the ordered pairs in a relation.
The **Range** is the set of all *second* numbers of the ordered pairs in a relation.

A **Function** is when none of the domain values are repeated.
A **Vertical Line Test** is a test for functions.

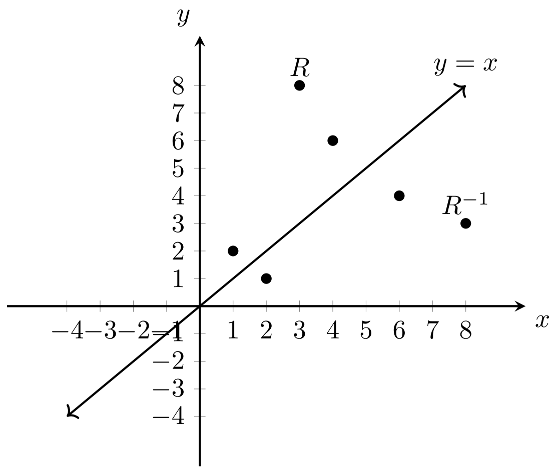
11.1.1 Compositions of Functions

A **composite function** is when a variable is replaced with another function.

Example: f(g(x)) is read as: 'f of g of x'

Inverse Relations is when you reverse ordered pairs of an initial relation.
The **identity function** is the line which if you folded the graph over

on this line/crease it would line up both relations. Since for each replacement of x, the result is identical to x.



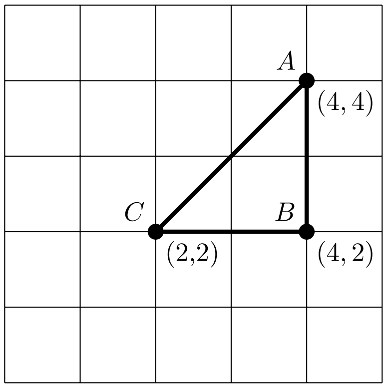
Chapter 12

Coordinate System

12.1 Rectangular Coordinate System

Coordinates of a point is how each point on a plane is assigned a pair of numbers.
Origin is the point of *intersection* of the x-axis and y-axis.
Coordinate Plane is the x-axis, y-axis, and all the points in their plane.
Ordered Pairs is a pair of numbers, written in parethesis and separated by comma for a point. Every point on a coordinate plane determined by order of numbers. example being (0,0) for the origin.
X-Coordinate is the number to the left of the comma in an ordered pair and determines the movement along the x-axis from origin. It moves right for positive and left for negative.
Y-Coordinate is the number to the right of the comma in an ordered pair and determines the movement perpendicular from the x-coordinate. It moves above the x-coordinate for positive and below the x-coordinate for negative.
Quadrants are the four separated regions of a coordinate plane. Starting on top right and going counter-clockwise they're are named Quadrant I-IV.
Graph is the point associated with an ordered pair of real numbers.

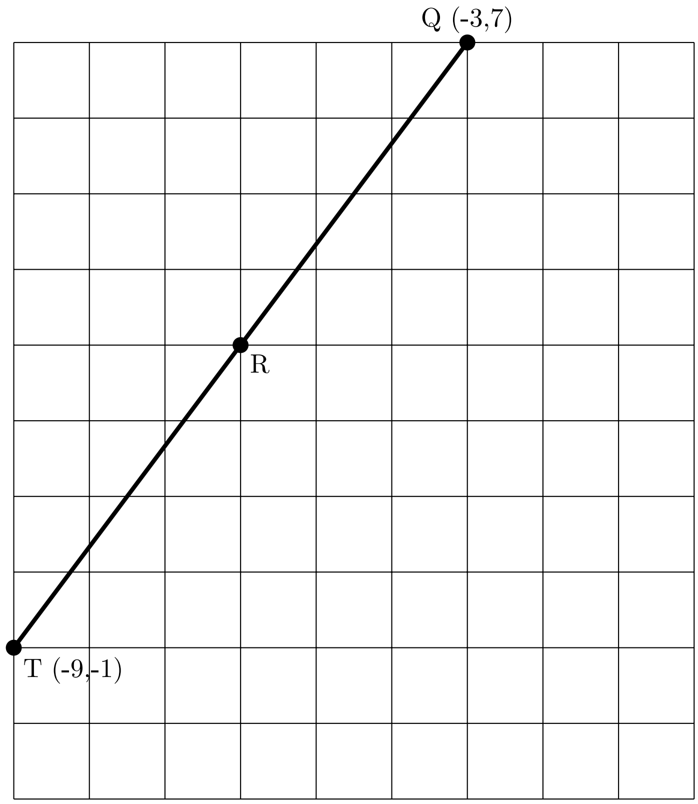
12.1.1 Distance Formula



$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2} \tag{12.1}$$

All distances can be found by subtracting, except the *hypotenuse*, which is found using the **Pythagorean Formula**.

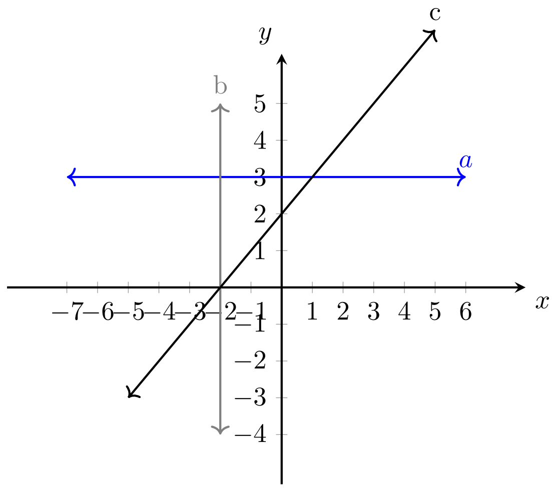
12.1.2 Midpoint Formula



$$M = (\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}) \tag{12.2}$$

Midpoint is the average of the endpoints.

12.1.3 Slope of a Line



$$\boxed{m = \frac{y_2 - y_1}{x_2 - x_1}}$$

The **slope of a line** is a measurement of steepness and direction of a *nonvertical* line. This means line **b** has an undefined slope, while slope **a** has a slope of 0. Meanwhile, **c** has an x-intercept at -2 and a y-intercept at 2. These intercepts are when either x or y are equal to 0.

12.1.4 Slopes of Parallel and Perpendicular Lines

Parallel lines have equal slopes and **Perpendicular** lines have *opposite* slopes/their product is -1.

Example: Given the original slope:

$$m = \frac{3}{4}$$

The slope of a parallel line ($m_{\|}$) is the same:

$$m_{\|} = \frac{3}{4}$$

The slope of a perpendicular line ($m_{\perp}$) is the negative reciprocal:

$$m_{\perp} = -\frac{4}{3}$$

12.1.5 Equations of Lines

1. If a point lies on the graph of an equation, then its coordinates make the equation true, and

2. If the coordinates of a point make an equation a true statement, then the point lies on the graph of the equation.

They can be written in the form of Ax + By = C, where A & B are not zero. If they were zero then C would always have to be zero too making the form false.

Standard Form is $Ax + By = C$ for the equation of a line.
X-Intercept is where the y-coordinate is 0.
Y-Intercept is where the x-coordinate is 0.

Note: Horizontal lines have no x-intercept unless its on the x-axis. Vertical lines have no y-intercept unless its on the y-axis.

12.1.6 Graphs of Linear Inequalities

Linear Inquality is a sentence in the form of $Ax + By < C$. This can be any less than, greater than, or less/greater than or equal to symbol.

If it is 'or equal to' then you make the line solid when graphing.
If it is not, then you make the line dotted as a boundary. This is because you are not counting what is on the line.

Now you want to plot a test point that is *not* on the line. If this test point is true for the inequality then shade the side it is on, if not, shade the opposite side if it is false.